

**Studies on dry root rot [*Rhizoctonia bataticola* (Taub.)
Butler] of chickpea**



THESIS

submitted to the

Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya, Gwalior

In partial fulfilment of the requirements for

The Degree of

MASTER OF SCIENCE

IN

AGRICULTURE

(PLANT PATHOLOGY)

BY

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2018

CERTIFICATE - I

This is to certify that the thesis entitled "**Studies on dry root rot [*Rhizoctonia bataticola* (Taub.) Butler] of chickpea**" submitted in partial fulfillment of the Degree of **MASTER OF SCIENCE IN AGRICULTURE (Plant Pathology)** of Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya, Gwalior is a record of the bonafide research work carried out by **Mr. Tinkulal Muchhala** under my guidance and supervision. The subject of the thesis has been approved by the Student's Advisory Committee and the Director of Instruction.

No part of the thesis has been submitted for any other degree or diploma or has been published. All the assistance and help received during the course of these investigations has been acknowledged by the scholar.

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This is to certify that the thesis entitled "**Studies on dry root rot [*Rhizoctonia bataticola* (Taub.) Butler] of chickpea**" submitted by **Mr. Tinkulal Muchhala** to the **Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya, Gwalior**, in partial fulfillment of the requirements for the degree of **Master of Science in AGRICULTURE** in the **Department of Plant Pathology** has been accepted after evaluation by the external examiner and approved by the Student's Advisory Committee after on oral examination the same.

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ACKNOWLEDGEMENT

*I praise the Almighty for enabling me to accomplish the task of this dissertation by His grace and for His glory. I find no words to express my sincere feelings of gratitude towards my esteemed guide and chairman of my advisory committee, **Dr. S.P. Mishra**, Professor, Hade, Department of Plant Pathology, who deserves my most sincere thanks and respect for inspiring and excellent guidance, contact, encouragement, unceasing interest, constructive criticism and helping attitude throughout the investigation and preparation of the manuscript.*

*I extend my sincere gratitude to other members of my advisory committee **Dr. R. K. Singh** Assistant Professor Department of Plant Pathology College of Agriculture, Indore and **Dr. R. K. Choudhary** Professor Department of Entomology, College of Agriculture, Indore and **Dr. Ashok Krishna**, Dean College of Horticulture, Mandsour for their suggestions and helping attitude during the research work.*

*With profound respect, I wish to express my sincere gratitude to **Dr. S.K. Rao**, Vice Chancellor RVSKVV Gwalior, **Dr. A. K. Singh** Director Instructions (RVSKVV), **Dr. U. R. Khandkar**, Dean, College of Agriculture, Indore, (M.P) for providing necessary facilities during the experiment.*

*I am also extremely thankful to **Shri Shankar Lal Choudhary**, Lab technician, Department of Plant Pathology.*

I thank to my senior Mukesh sir, Pankaj sir, Jitendra sir and my all my friends, Imlesh Birla, Pradeep, Rahul Aske, Pooja, Bhavana, Bandana, Anil Birla, who helped me directly or indirectly during the course of investigation.

I find no rhetorical gems from the ocean of word to express my pro-found feeling to my venerable parents Mr. Lakhn Lal Muchhala and Mrs. Sita Muchhala who were always present in all of my good and the bad times. I also want to express my profound feelings to My loving sister Deepika and my brathore Mr. Hemant Muchhala, Kamlesh Muchhala and Akhilesh, who have piloted me up to this stage and whose love, devotion, blessings and care throughout my life enabled me to achieve this seemingly invincible goal.

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ABBREVIATIONS

S.N.	Legends	Description
1	@	At the rate of
2	%	Per cent
3	⁰ C	Degrees Celsius
4	ANOVA	Analysis of variance
5	C.V.	Coefficient of variation
6	cal.	Calculated
7	CD	Critical difference
8	Cfu	Colony forming unit
9	Cm	Centimeter
10	CRD	Completely Randomized Design
11	DAT	Day after transplanting
12	DF	Degrees of freedom
13	EC	Emulsifiable Concentrate
14	EMS	Error means sum of square
15	<i>et al.</i>	Allied (and other)
16	Fig.	Figure
17	G	Gram
18	g ^l ⁻¹	grams/ liter
19	i.e.	That is (in reference to)
20	Kg	Kilogram
21	mm.	Millimeter
22	MSS	Mean Sum of Square
23	No.	Number
24	PDA	Potato Dextrose Agar
25	Ppm	Parts Per Million
26	Q	Quintal
27	S.N.	Serial number
28	S.S.	Sum of Square
29	S.V.	Source of variance
30	SE (m)±	Standard error of mean
31	SL	Suspendable liquid
32	spp. & sp.	Species
33	tab.	Tabulated
34	t/ha	Tones/hectare
35	Viz.,	Wide list
36	WG	Wettable granule
37	WS	Water Soluble

CHAPTER-I INTRODUCTION

Pulses constitute major sources of proteins among the vegetarians in India and complement the staple cereals in the diets with proteins, essential amino acids, vitamins (A, B and C) and minerals. They contain 22-24% protein, which is almost twice the protein in wheat and thrice that of rice. Pulses provide significant nutritional and health benefits, and are known to reduce several non-communicable diseases such as colon cancer and cardiovascular diseases (Jukanti *et al.*, 2012).

Chickpea is an important source of protein for millions of people in the developing countries, particularly South Asia as the people are largely vegetarian either by choice or because of economic reasons. In addition to having high protein content (23%), chickpea is rich in, carbohydrates (47%), starch, (6%), soluble sugars, fat (5%), crude fibre (6%), ash (3%) and β -carotene (Gaur *et al.*, 2010). Its lipid fraction is high in unsaturated fatty acids. Since chickpea is a rich and cheap source of protein in comparison with animal protein, it is called as the 'poor man's meat'.

Chickpea plays a significant role in improving soil fertility by fixing atmospheric nitrogen. It meets 80 per cent of nitrogen (N) requirement from symbiotic nitrogen fixation and can fix up to 140 Kg N ha⁻¹ from air and leaves substantial amount of residual nitrogen for subsequent crops and adds plenty of organic matter to soil health and improve the health and fertility. Because of its deep tap root system, chickpea can withstand drought conditions by extracting water from deeper layers in the soil profile (Gaur *et al.*, 2010).

India contributes the major share to the global chickpea area and production (both about 65%). In India, it is grown on 9.93 mha area with a production of 9.53 mt and productivity of 960 kg/ha (Anonymous, 2015). Madhya Pradesh, Uttar Pradesh, Rajasthan, Maharashtra, Karnataka, Bihar, Andhra Pradesh, Chhattisgarh, Haryana, West Bengal are the Chickpea growing areas of India. Among the chickpea growing states, Madhya Pradesh is one of major chickpea growing state in India, which covers the largest area of 3.16 mha and contributes about 3.29 mt with an average productivity of 1044 kg /ha (Anonymous, 2015).

Chickpea productivity often remains low due to susceptibility of the crop to different biotic and abiotic stresses. Among the biotic stresses, diseases, insect pests,

nematodes and parasitic weeds account for major losses. The crop is attacked by 172 pathogens which include 67 fungi, 22 viruses, 3 bacteria, 80 nematodes and phytoplasma from all over the world (Nene *et al.*, 1984).

Among the diseases dry root rot caused by *Rhizoctonia bataticola* (Taub.) Butler (pycnidial stage: *Macrophomina phaseolina* Tassi Goid) is emerging as serious threat to the chickpea growing regions of India (Masood and Shivkumar, 2001). The pathogen survives in soil on infected host debris. Secondary spread of the disease is through farm implements, irrigation water and rain splashes. Recently wilt resistant chickpea cultivars were found succumbing to dry root rot in major growing areas in India (Pande *et al.*, 2010). Being soil borne in nature, it is very difficult to control this pathogen. Yield losses due to root rot are more when the conditions are congenial for disease development. Losses to the extent of 70.8 per cent at full podding and 48.9 per cent at preharvest stage of the crop have been reported due to the disease (Ahmed and Mohamad, 1986).

R. bataticola is a diverse, omnipresent soil borne fungal pathogen with a wide host range involving more than 500 species in 75 families (Dhingra and Sinclair, 1978). It mainly produces micro-sclerotia (primary source of inoculum) or pycnidia. It is both soil and seed borne and infects plants mainly from flowering stage to maturity. The disease is reported to be more severe when chickpea crop is exposed to moisture stress conditions (Sharma *et al.*, 2010). Natural population of this pathogen is diverse in terms of pathogenic variability and can be characterized into pathotypes or races by means of biological typing. Previous studies have revealed and reported high levels of pathogenic and genetic variation in *R. bataticola* from different parts of the world and India (Trivedi and Gurha, 2006; Aghakhani and Dubey, 2009; Pande *et al.*, 2010).

Prolonged saprophytic survival ability of the pathogen in soil makes chemical control and crop rotation ineffective. Cultivation of resistant varieties is an economical approach for the management of dry root rot of chickpea, but only a few sources with low level of genetic resistance are available.

Management can be made feasible and cost effective by identification of new resistant sources for which development of sick plot is essential which is the only alternative method so as to combat this serious disease, even though chemical control plays an important role in modern agriculture for disease management (Gupta *et al.*, 1997).

Environmental conditions like temperature, soil moisture and pH play an important role in the viability and growth of *Rhizoctonia bataticola* (Khan, 2007). *R. bataticola* is able to produce microsclerotia under relatively low water conditions while viability of microsclerotia is drastically reduced at high water potentials (Olaya *et al.*, 1996). Documenting the

independent and interactive effects of soil moisture on plant disease is essential for developing effective management strategies or screening for host resistance. Keeping this in view, present investigations were envisaged with the following objectives.

1. To screen the advanced chickpea lines for resistance to dry root rot in the sick soil.
2. To study the influence of soil moisture and organic matter on development of dry root rot.
3. To evaluate the techniques for artificial inoculation of dry root rot.

CHAPTER II

REVIEW OF LITERATURE

Chickpea is most extensively grown pulse crop in India. It is attacked by air borne, seed borne, and soil borne pathogens. More than 75 pathogen have been reported to infect chickpea (Nene *et al.*, 1984).

Among soil borne diseases dry root rot (*Rhizoctonia bataticola* Taub. Butler) is most important diseases causing heavy losses from flowering to podding stage and disease severity depends upon the soil moisture condition/regimes.

2.1 Historical

The first mention of a plant disease, which may be referred with certainly to *Rhizoctonia* as a causal agent is an important paper by Du Hamel (1728) read before the Paris Academy. In this paper he gave a careful description of a fungus disease of saffron (*Crocus sativus*) occurring in France. He named the fungus *Tuberorides*.

De Candolle (1815) discovered a similar disease on alfalfa. In his taxonomic description he was the first person who gave the name for the fungus. Later, after careful study of the characteristics and parasitism of the species, he established the genus *Rhizoctonia* in 1815.

In 1924, small described a *Rhizoctonia*, which develop black sclerotia of variable size and shape in the diseased roots, and many woody plants from Uganda and named it *R. lamrllifera*. Butler (1925), however considered that the fungus should be regarded as *Rhizoctonia* and made the combination *R. bataticola* (Taub.) Butler to a *Sclerotium* causing a root rot of cotton, french beans and cowpeas.

2.2 Pathogen

Rhizoctonia is a genus of anamorphic fungi in the order Cantharellales and family Ceratobasidiaceae. *Rhizoctonia* species do not produce spores, but are composed of hyphae and sclerotia (hyphal propagules), asexual stage of fungi. *Rhizoctonia* species are saprophytic, but some act as facultative plant pathogens causing commercially important crop diseases. The genus *Rhizoctonia* (means “root killer”) was described in 1815 by the French mycologist Augustin Pyramus de

Candolle for plant pathogenic fungi that produce both hyphae and sclerotia. Subsequent authors added over 100 additional names to the genus. A comprehensive survey and redispotion of species name "Rhizoctonia" was published in 1994 by Anderson and Stalpus. According to this published information, *R. bataticola* is used as a synonym to *Macrophomina phaseolina* (tassi) Goidanich, 1947. Currently, *M. phaseolina* is officially recognized as the correct taxonomic name. The sclerotial phase known as *R. bataticola* (Holliday and Punithalingam 1970).

The species description of *R. bataticola* is given by Subramaniam (1971). "The basic characteristics of the species are the formation of sclerotia in the host tissue as well as in culture on the infected plant sclerotia seen in the pith region and outer surface of root just below the bark region. They are brown or jet black, minute, smooth externally, hard, variable in shape, globose, oval oblong, elliptical, curved or even forked and varying in size, but of uniform texture more or less loosely packed." The cell of hyphae are barrel shape, branching more or less at right angles and light to dark brown in colour composed of free thick walled cell. The species *Rhizoctonia* is distinguished by light brown to brown hyphae with somewhat loose/ spongy sclerotia in the form of bulbous.

Mycelia width varied from approximately 2-11µm and distance between two consecutive septa measured 46µm. However, the most important character regarding taxonomy and classification were the production size and composition of microsclerotia (Reichert and Hellinger, 1947).

Sclerotia within plant parts were black, smooth, hard and varied in size from 100 µm-1mm while in culture, it varied from 50-300µm. These descriptions were given by Commonwealth Mycological Institute (CMI), Kew, England (1970). During the sclerotial formation, 50–200 individual hyphal cells aggregate to give multicellular bodies called microsclerotia. The microsclerotia were black and variable in size from 50–150 µm depending on the available nutrients of the substrate on which the propagules were produced (Short and Wyllie, 1978).

In vitro conditions, the fungus grows rapidly on potato dextrose agar (PDA) and produces brown to grey coloured mycelium that become darker with age. The young hyphae are thin, hyaline, aseptate and dichotomously branched and later

produce typical black sclerotia. The fungus may produce abundant aerial mycelium, as high as to touch the cover of the culture plates, or mycelium is found to be completely or partially suppressed (Sharma *et al.*, 2012). The characteristic features of *R. bataticola* are right angle branching of the mycelium and constriction of the branch near the point of origin. The sclerotia formed are black, smooth, varying from spherical through oblong to irregular shapes (Sharma *et al.* 2012)

2.3 Occurrence and distribution of disease

The first report of the occurrence of root rot in chickpea along with wilt was made by Padwick (1948). The incidence of root rot ranging from 3.58 to 20.63 per cent in 30 villages of northern Madhya Pradesh was reported by Gupta *et al.*, (1983).

Dry root rot chickpea was first reported from India by Mitra (1931), later, the disease has been reported from most chickpea-growing areas in India like Punjab by Luthra (1938), Madhya Pradesh (Kotasthane *et al.*, 1979), Uttar Pradesh (Pandey and Singh, 1990). It has been reported from other countries like Iran (Kaiser *et al.*, 1968), the USA (Westerlund *et al.*, 1974) and several countries in Asia and Africa (Nene *et al.*, 1996). The disease was earlier known as “*Rhizoctonia* wilt” in chickpea; however, later it was named as “dry root rot”.

R. bataticola is responsible for serious diseases like stem blight, foliage blight, root rot, seedling blight, charcoal rot, wilt, collar rot, dry rot, pod and seed rot in several crops such as soybean, gram mung, urid, fruit trees and many vegetables (Mukharji and Bhasin, 1986). This *sclerotial* fungus is of wide geographical distribution ranging from Formosa, Philippines, India, Ceylon, East Africa, Palestine and Egypt to the West Indies and the Eastern United State (Ashby, 1927). Significant reduction (25-70%) in chickpea production is caused by this disease in India (Ahmed *et al.*, 1986 and Pandey and Singh, 1990).

Nema and Khare (1973) reported that chickpea cultivation in Madhya Pradesh was greatly threatened by soil borne disease generally appearing at two phase of growth early seedling stage and flowering stage wilting was due to *Fusarium oxysporium* f. sp. ciceri, Dry root rot caused by *R. bataticola* and *Sclerotium rolfsii* resulted in collar rot.

Tripathi and Sharma (1983) studied the incidence of chickpea dry root rot in Southern Haryana and reported that the incidence was high from late October to mid

November which was decreased in December and January and then again increased in the months of February and March. They also noted that seed sown at a depth of 15 cm developed a less root rot than that sown at 7.5 cm.

Shrivastava *et al.* (2002) conducted a survey in Vindhyan Plateaus zone of Madhya Pradesh India to evaluate the prevalence of wilt and dry root rot disease in chickpea. They reported that *F. oxysporum* was most prevalent (56%) in wilted plant followed by *R. bataticola* (17%) and *Trichoderma sp.* (10%). The incidence of *R. bataticola* also caused withering and drying of plant, which range from 8 to 20 percent.

Manjunatha *et al.* (2011) reported maximum dry root rot incidence in chickpea in Gulbarga district (9.8%) followed by Raichur (7.6%) and the least (6.18 %) in Bidar (Karnataka). The overall incidence of dry root rot ranged from 1-19% across the districts. This was mainly attributed due to variation in soil type (black and red soil) and cultivars grown (Annegiri and JG 11) under rain fed condition.

Studies conducted by Jagre (2012) on succession of chickpea diseases stated that highest percent mortality due to dry root rot was observed in the 10th standard week i.e. 2nd week of march 2012 when the temperature range from 11.6-30.8 0c with 18-76 per cent relative humidity however the disease initiated when the temperature was > 27 0c with the decreases in relative humidity.

The dry root rot of was not of much significance in chickpea earlier, but now, it has become a major threat to chickpea production in recent years due to altered weather conditions, particularly on the account of longer drought spells. Higher temperature and soil moisture depletion during crop growth period particularly at post-harvesting stage is predisposing chickpea to dry root rot (Sharma & Pande, 2013).

2.4 Symptoms

R. bataticola is a soil borne fungal pathogen. Several workers have reported various types of symptoms produced on plants with varying intensity of the disease in different crops and different regions.

Narsimhan (1929) observed that plants infected with *R. bataticola* initially showed drooping of leaves followed by wilting. The main roots of such plants show blackening of finer roots.

According to Mehrotra and Aneja (1990) the first symptoms of the disease appeared as yellowing of the leaves within a day or two such leaves drop and in the course of the next two or three days whole plant dies off. The leaves and stem of the affected plants are usually straw coloured. The tap root is dark and quite brittle in dry soil.

Singh *et al.* (1990) observed the root inoculated with *R. bataticola* (Madhya Pradesh) and reported disintegration of cortical tissue while mycelium and sclerotial bodies plugging the xylem were seen in longitudinal section.

The tap root turns black with sign of rotting and is devoid of most of the lateral and tiner roots. The dead root are quit brittle and show shredding of the bark. The tip of the root is easily broken leaving the lower portion of the tap root in the soil, when plants are uprooted. Dark minute sclerotial bodies can be seen on the roots exposed and inner side of the bark or when split open at the collar region vertically (Sharma *et al.*, 2015).

2.5 Pathogenicity

Byadgi and Hedge (1985) studied the variation in size and shape of pycnidial and pycnidiospore of *M. phaseolina* on chickpea based on morphological, cultural character and pathogenic behavior.

Demiro *et al.*, (1994) observed that dry root rot disease of gram is also becoming a serious disease affecting productivity of this crop. They isolated *R. bataticola* from plants and seeds of chickpea cultivars Azizive 94 and tested pathogenicity.

Katariya *et al.* (2012) made studies to determine the cultural and pathogenic variation of *M. phaseolina* isolates collected from infected chickpea from various parts of Rajasthan. A total of 9 *R. bataticola* isolates, 7 from different districts of Rajasthan and one each from CCS, Haryana Agricultural University, Hisar and Indian Agricultural Research Institute, New Delhi, were isolated from infected chickpea samples. Pure cultures of these isolates were obtained and the pathogenicity of these isolates was tested on 40-day-old chickpea cv. C-235 grown in pots.

Amrutha *et al.* (2014) isolated *R. bataticola* from chickpea and identified the pathogen based on its mycelial and sclerotial characters and pathogenicity test was proved by soil inoculation method on susceptible variety.

2.6 Screening against the pathogen

Ahmed *et al.*, (1990) screened 211 promising ICRISAT lines against wilt and root rots mainly (*R. bataticola*), each entry and the susceptible check were grown in 4 m long rows, 30 cm apart in 1987 and 1988 and wilt/root rot incidence was recorded at 15 day intervals. The mortality of the susceptible control in 1986-87 main season, 1987 off season, 1987-88 and 1988-89 was 50%, 80%, 95% and 95%, respectively, indicating the uniformity of inoculum. Isolations from dead plants indicated that mortality was mainly due to wilt and dry rot. Among 48 entries which showed less than 20% mortality in the 1987-88 season, 46 lines again showed less than 20% mortality in the 1988-89 season; 22 lines, of which showed less than 10% mortality, were selected for an initial yield trial in 1989-90.

Baker and Ahmed (1991) reported additional sources of resistance to wilt and root rot of chickpea in Bangladesh, 12 were finally selected as resistance. Of these, ICC 12263 was most resistant, with a mean mortality rate of 2.3 percent, with other entries exhibiting mean mortality rates of up to 5 percent.

Gangwar *et al.*, (2002) screened 35 chickpea cultivars for resistance to dry root rot caused by *Rhizoctonia bataticola* in a field experiment. All the cultivars exhibited disease symptoms, only 10 cultivars (ICC 2644, 10384, 10630, 112244, 11332, ICCL 81002, 810810, ICC 12263, 12441 and ICCV 90254) were resistant whereas 5 genotypes (GCP 9504, Phule G 96020, 96105, 96313 and GL 91059) were moderately resistant to dry root rot.

Pande, *et al.* (2004) screened 29 chickpea germplasms accessions and 10 cultivars for resistance to dry root rot to confirm their resistance. Artificially inoculated 8-day-old seedlings were incubated at 35°C with 12 h photoperiod.

The dry root rot was scored on a 1-9 rating scale on the 8th day after inoculation. Based on the disease score (DS), the accessions were grouped as immune (DS=1), resistant (DS>1 and ≤3), moderately resistant (DS>3 and ≤5), susceptible (DS>5 and ≤7) and highly susceptible (DS>7). Of the

47 lines, none was immune to dry root rot. One germplasm accession (ICC 14395), a cultivar (ICCV 2) and an advanced breeding line (ICCX830203-BHBH-11H) were resistant to dry root rot. Of the remaining lines, 22 were moderately resistant, 19

susceptible and 3 highly susceptible. The disease severity in the 2 susceptible lines (BG 212 and ICC 12267) used as the controls was rated 9.

Gupta and Babbar (2006) screened 233 *desi* and *kabuli* chickpea genotypes along with 38 promising lines against dry root rot in multiple disease sick plot in Jabalpur, (M.P.) during 2001-03. Genotypes H 99-264, PG 9425-5 and PG 9425-9 (*desi*) and HK 00297 and PG 97-313 (*kabuli*) exhibited resistance against all the 3 pathogens. WCG 97-29, WCG 97-37, IPC 97-72 and H 00-256 (*desi*) showed less than 10% mortality for collar and dry root rot. GF 88240, MPJG 98-9032 and MPJG 98-11551 (*desi*) and HK 94-134 (*kabuli*) showed resistance to wilt and dry root rot disease. These genotypes are valuable sources of resistance and may be exploited for utilization in breeding programmes for developing high-yielding cultivars with multiple disease resistance.

Vijay *et al.* (2006) evaluated twelve chickpea cultivars for resistance against dry root rot disease under natural and artificial conditions. Screenings under natural conditions were conducted during the rabi seasons of 2001-02 and 2002-03 in Jharkhand, India. For artificial conditions, potted soils were inoculated with mycelial bits of the pathogen and seeds were sown at 4 seeds per pot three days after soil inoculation. Under natural conditions, the mean disease incidence was lowest (8.0%) in G-543, followed by BG-360 and GNG-1365 which recorded mean disease incidence of 9.6 and 9.8%, respectively. These cultivars showed resistant reaction against dry root rot disease. Pusa-372, RSG-888 and RSG-931 showed moderately resistant reaction against the disease. Under artificial conditions, all the cultivars were infected. The mean disease incidence ranged between 14.3 and 41.8%. Cultivars G-543, BG-360 and GNG-1365 showed moderately resistant reaction. Pusa-372, GNG-1292, RSG-888, RSG-895, H-208, CSJ-103 and RSG-931 showed susceptible reaction against the disease. G-543, BG-360 and GNG-1365 showed resistant reaction under natural conditions and moderately resistant reaction under artificial conditions.

Jayalakshmi *et al.* (2008) conducted a pot culture experiment for screening of chickpea entries (12+1 susceptible check JG-62) during *rabi* season of 2004-2005 and 2005-2006. Out of 12 genotypes tested against root rot diseases four were

found resistant, two were moderately susceptible, three were susceptible and 3 were highly susceptible to dry root rot disease.

Gupta *et al.* (2012) screened for resistance against *Rhizoctonia bataticola* causing dry root rot in chickpea. The susceptible cultivar BG 212 showed 100 per cent mortality *i.e.* rating 9 in 1-9 scale. The studies led to conclusion that out of 170 accessions, 68 genotypes exhibited resistant reaction (<10% mortality), out of which 26 are the promising lines namely JG1- 14, 2-125, 2-4-110, 14-11, 14-10, 2001-13, 2001-13, 2001-18, 2001-80, 2001- 115, 2002-20, 2003-95, 2003-14-16, 2004-110, 210,9605, 1-9, 99-115, 2001- 04, 2003-14-2, JG 2000-07, JSC 37, MPJG89-11551, MPJG 89-9023, CSJ 592 and Rajas from JNKVV, Jabalpur. These lines further evaluated for their performance in sick field for three consecutive years and revealed six lines *viz.*, JG 2000-07, JSC 37, MPJG 89-11551, MPJG 89-9023, CSJ 592 and Rajas as resistant exhibiting <10 per cent mortality, however 14 lines showed moderately resistance reaction.

Indu *et al.* (2012) screened 25 genotypes of chickpea including three checks under natural epiphytotic conditions for their reaction against wilt/root rot complex diseases in two different environments.

Lekhraj *et al.* (2012) found pathogenic variability on chickpea infected with *M. phaseolina* isolates collected from 9 various parts of Rajasthan, Hissar and New Delhi. Pathogenicity of these isolates on 40 day old chickpea cv. C-235 showed no direct relationship between cultural characters and virulence of the isolates. The isolate BKN had scanty, suppressed growth and was highly pathogenic. Isolates SGN, NHR, DUR, HAU, DEL and HNG having different types of colony growth were moderately pathogenic and the isolates CHR and UPR with different mycelium characters were less pathogenic. Similarly, there was no apparent correlation between the virulence and growth rate of the fungal colonies.

Fifty isolates of *R. bataticola* collected on chickpea from different agro climatic zones in India proved pathogenic on chickpea cultivar BG 212. The dry root rot disease severity ranged from 4 to 9 (1-9 severity rating scale). The effect of isolates on severity was significant. The pathogenic and non pathogenic isolates were not concentrated in any one particular state/region (Sharma *et al.* 2012).

Among the 94 isolates of *R. bataticola* collected from different agro ecological zones, 8 isolates from Andhra Pradesh and Madhya Pradesh showed less aggressive on chickpea cultivar BG 212 (Sharma *et al.*, 2012).

2.7 Effect of moisture on disease development

Raabe (1985) found that *R. bataticola* causes severe damage to chickpea in the warmer Salinas Valley in California as the pathogen is becoming more virulent under high temperatures (32°C). *R. bataticola* can grow and produce large amounts of microsclerotia under relatively low water potentials allowing the disease on bean to be recognised as favouring drought (Olaya and Abawi, 1996). The survival of sclerotia of *R. bataticola* in black soil was affected by high soil moisture (80%), while the saprophytic activity of pathogen was maximum at low moisture levels (20,30 and 40%) (Maheswari and Ramakrishnan, 1999). The adverse effect of high soil moisture on survival of sclerotia of *R. bataticola* was also reported by Olaya and Abawi, (1996) indicating that *R. bataticola* cannot survive for a longer period under anaerobic conditions. Similar to chickpea, in soybean and cowpea, *M. phaseolina* population in the soil was negatively correlated with soil moisture and positively with maximum soil temperature. (Gupta and Gupta, 1986). The dry root rot incidence of chickpea was maximum in sandy soils (86.6%) as compared with clay soils (61.6%) as observed by Taya *et al.* (1988) in sick pots at greenhouse. The deficit soil moisture was generally detrimental to both chickpea plant growth and pathogen activity (Bhatti and Kraft, 1992).

Dry root rot was observed in gram when exposed to temperature more than 30°C. The disease was aggravated by dry soil conditions especially at flowering stage leading plants to sudden drying (Singh and Mehrotra, 1982).

Colonization of *M. phaseolina* on chickpea roots was high when subjected to moisture stress conditions compared to unstressed plants. Soils artificially inoculated produced poor growth of plants both in low and high moisture compared to natural soil. Root colonization was greater at 40-50% moisture holding capacity as compared to 10-20% soil moisture holding capacity (Husain and Ghaffar, 1995).

Ratnoo *et al.* (1997) reported the infection and development of ashy grey stem blight in cowpea caused by *M. phaseolina* was most favoured by higher temperature 25-40°C and low in flooded soil compared to dry soil of 40-60% soil moisture.

Role of edaphic factors on the incidence of dry root rot of sesame caused by *R. bataticola* were studied by Raj Krishan *et al.* (1999). Among various soil types, sandy soil supported maximum disease incidence (78.33%) as compared with clay soil (51.56%). More disease incidence of 48.33% was observed when pots were irrigated after the interval of seven days as compared to pots receiving irrigation every day.

Wokocha (2000) observed the disease incidence of dry root rot of soybean was significantly low when the soil moisture was 60 - 70% but it was 5.0% at 10 - 20% soil moisture

Patel and Anahosur (2001) conducted an experiment with chickpea cultivars ICC 4951 (susceptible) and Bheema (resistant) in earthen pots filled up with artificially infected soil with *M. phaseolina* at four levels of soil moisture viz., 25, 50, 75 and 100%. They observed *M. phaseolina* infection was maximum at 25% and minimum at 100% soil moisture. ICC 4951 developed higher disease than Bheema at 25% soil moisture. Both cultivars showed maximum disease incidence at the lowest soil moisture level compared to high moisture.

Singh and Sharma (2002) observed that deficit soil moisture favors the severe disease development on pulse crops. Under hot and dry environment many economically important crops are predisposed to the infection and colonization of *R. bataticola* and caused drastic yield losses on chickpea (Thripathi and Sharma, 1983), soybean (Pearson *et al.*, 1984) and sunflower (Nawaz, 2007).

With increasing temperature and more frequent moisture stress, dry root rot is becoming more intense in typically tropical humid areas. Increasing incidence of dry root rot at various locations over years suggests a strong influence of rising temperatures on *R. bataticola* (Savary *et al.*, 2011).

Role of soil moisture on disease severity of dry root rot in chickpea was studied by Sharma and Pande (2013) under controlled environment. The plants exposed to 40% moisture stress showed higher mortality as compared to 60%, 80% and 100%. It was observed that 40% soil moisture was insufficient for the normal growth of the plants as the plants grown in control showed the physiological stress (wilting of the plants due to lack of moisture). At 60% soil moisture, no physiological stress was found in control plants and dry root rot severity was maximum 35 days after imposing

moisture stress. Disease progressed slowly at 80% and 100% soil moisture had least disease severity.

CHAPTER-III

MATERIAL & METHODS

The present investigation on dry root rot of chickpea involving the aspects of screening of available lines/germplasms against the pathogen, effect of soil moisture on development of dry root rot and standardization of technique for artificial inoculation were carried out in the laboratory and in the field during Rabi 2017-18 in the Department of Plant Pathology, College of Agriculture, Indore (M.P.). The materials used and the methods followed are being described in this chapter.

3.1 Materials

The materials used included seeds of twenty chickpea lines and isolate of *R. bataticola*, seeds of chickpea variety L-550, susceptible to dry root rot, chemicals for culture media, glass wares, equipments and a few miscellaneous articles.

3.1.1 Source of seeds

Chickpea seeds were procured from AICRP (All India Coordinated Research Project) on Chickpea, Indore (M. P.).

3.1.2 Chemicals

(a) Cleaning solution

Cleaning solution contained potassium dichromate 80 g, distilled water 300 ml and concentrated sulphuric acid 400 ml. This was used to clean the glass wares.

(b) Mercuric chloride solution

A 1:1000 solution of HgCl_2 was prepared and used for surface sterilization of samples.

(c) Mounting medium

Lactophenol and cotton blue with the following composition were used as the staining medium for studying the characteristics of the pathogen.

(d) Lacto phenol

(Phenol pure crystals, liquefied 20 g by gentle heating on a water bath)

Lactic acid	20 ml
Glycerol	40 ml
Distilled water	20 ml

(e) Cotton blue

Anhydrous lacto phenol	67 ml
Distilled water	20 ml
Cotton blue	0.1 g

3.1.3 Glasswares

Standard "Borosil" make glasswares like Petri dishes, beakers, *funnels*, pipettes, Erlenmeyer flasks, culture tubes, measuring cylinder etc. were used during the course of study.

3.1.4 Equipments and miscellaneous articles

Equipments used during course of investigation included research Binocular microscope, refrigerator, autoclave, hot air oven, BOD incubator, Laminar, air flow, weighing balance, LPG gas burner and hot plate. Small instruments like inoculation needle, scalper, razor, glass cavity slides, glass rod, cork borer, wash bottle, forceps, scissor, knife, polypropylene bags,

desiccators, glass marking pencils, cover slips, brush, dropper, match box, plastic pots etc were also used during the study.

3.1.5 Composition and preparation of different media

(a) Potato dextrose agar medium

Potato dextrose agar (PDA) medium having the following ingredients was used for physiological studies.

Peeled and sliced potato (par boiled extract)	200 g
Dextrose	20 g
Agar -agar	20 g
Distilled water	1l.
pH (adjusted to)	6.5

3. 2 Methods

3.2.1 Cleaning and sterilization of glassware

The glasswares were cleaned by dipping them in cleaning solution for 5 minutes and finally rinsed with running tap water for 30 minutes. The Petri dishes were sterilized in a hot air oven at $180 \pm 2^{\circ}\text{C}$ for 1.5 to 2 hours. The inoculation needle, cork borer and other metallic instruments were sterilized by dipping them in alcohol and heating red hot over flame of the Bunsen burner.

(a) Cleaning of glassware

Borosil and Corning glasswares were kept in the cleaning solution containing 60 g of potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) and 60 ml of concentrated sulphuric acid (H_2SO_4) in one litre of water for whole day. Then they were cleaned by washing with

detergent powder followed by rinsing several times in tap water and finally in distilled water.

(b) Sterilization of The medium

The culture medium and water were sterilized in an autoclave at 1.1 kg per sq cm pressure for 25 minutes.

3.2.2 Surface sterilization of diseased specimens

Plant materials were surface sterilized using 0.1 per cent mercuric chloride solution for 1 minute and then washing in sterile water thrice to remove isolates of mercuric chloride.

3.2.3 Collection of diseased samples

Chickpea plants showing typical symptoms of dry root rot that is, dead roots were quite brittle and show shredding of bark and lateral root, at flowering and pod formation stage were collected from experimental field of department of Plant Pathology, Indore (M.P.).

3.2.4 Isolation of pathogen from diseased specimens

The roots of diseased plant showing symptoms were washed thoroughly with tap water, small pieces of infected roots were cut with the help of sterilized blade. These pieces were surface sterilized with 1:1000 mercuric chloride (HgCl_2) solution for one minute followed by three washings with sterilized distilled water to remove traces of HgCl_2 . The pieces were then transferred aseptically to Petri plates containing PDA. Inoculated Petri plates were incubated at $25 \pm 2^\circ\text{C}$ for three to five days and examined at frequent intervals to see the growth of the fungus developing from different pieces. As and when fungal colony appears they were transferred to PDA slants.

3.2.5 Purification and identification of isolates of *R. bataticola*

The cultures of *R. bataticola* were purified by frequent sub culturing by the hyphal tip method and maintained by mass transfer on potato dextrose agar medium at $25 \pm 2^\circ\text{C}$. These isolates of *R. bataticola* were identified by observing

the colony character, pattern of sclerotial formation and sclerotial size measured with the help of a micrometer.

3.2.6 Screening of lines for resistance to dry root rot in the field

Prepared mass culture of isolates of dry root rot pathogen was used to develop dry root rot sick bed in an augmented design. Twenty lines/cultivars of chickpea obtained from Department of Plant Pathology, College of Agriculture, Indore were screened for the sources of resistance against dry root rot chickpea. Each of the test line was sown in 3 meter paired row spacing with 60cm distance was raised following general agronomic practices. The data on the number of dry root rotted plants in each test line were recorded at 15 days interval. Observations on per cent mortality were started from ten days after inoculation and recorded at five day intervals up to maturity, finally computed as follows.

Table-3.1: IIPR rating scale

S. No.	Reaction	Per cent mortality
1	R- Resistant	0-10
2	MR- Moderately Resistant	11-20
3	MS- Moderately Susceptible	21-30
4	S- Susceptible	31-50
5	HS- Highly Susceptible	51-100

The reaction of all the entries as disease incidence was calculated by the following formula.

$$\text{Incidence (\%)} = \frac{\text{Total number of observed plants}}{\text{Number of diseased plants}} \times 100$$



**Seed Inoculation
Technique**



**Pot Inoculation
Technique**



**Drenching of
mycelium
suspension**



**Root and Fine root
cut mycelia suspension
dip technique**

**Plate-1: Techniques for artificial inoculation
of dry root rot**

3.2.7 Effect of soil moisture on chickpea dry root rot

Influence of soil moisture was examined on the dry root rot of chickpea in the field. Disease incidence was calculated weekly and soil moisture also observed weekly. Soil moisture recorded from 8-10 places by hot-air oven drying method. Soil samples were collected in aluminium container. Before the sampling, empty container (heat resistant) including cover is weighed (M_1), after that soil sample was collected from the field and soil was placed in the container and weighed together with the container (M_2) and place the containers in to hot air oven at 60 °C for 2 hour. After cooling of containers, soil plus container are weighed again (M_3). The moisture per cent was calculated by the following formula

$$\text{Moisture \%} = \frac{(M_2 - M_3) \times 100}{(M_2 - M_1)}$$

The data on disease incidence were plotted against soil moisture to predict the simple correlation and multiple correlation between disease and soil moisture. The significance of correlation coefficient and multiple correlation were judged by 't' test and 'f' test respectively, as per standard statistical methods.

Location: The experiment was conducted in Research field, College of Agriculture, Indore (M.P.)

3.2.8 Evaluation of the techniques for artificial inoculation of dry root rot

An experiment was conducted to identify the inoculation technique for establishment and development of dry root rot by the following inoculation methods (Plate-1).

3.2.8.1 Seed Inoculation Technique

Healthy surface sterilized seeds of highly contaminated seed sample were taken. The seeds were then rolled, separately on 10 days old sporulating culture of each root rot causing fungi thriving on PDA in Petri-dishes. The inoculated seeds were plated on moist blotter papers in Petri dishes. Ten seeds were plated at nearly equal distance in each Petri-dish. After plating, Petri-dishes were incubated.

3.2.8.2 Pot inoculation method

The test fungus was multiplied on soil maize medium for 7 days. Sterilized soil was used in pot experiment. The soil was collected from the local field and dried thoroughly in sun and sterilized. Pots of 25 cm face diameter were used during experiments.

The pots were filled with 1kg sterilized soil and then 20g of inoculum was added in each pot. One inch of soil was placed over the inoculum. The contents were then mixed and surface sterilized seeds were sown and covered by half inch layer of soil. In control, no inoculum was added and pots were filled with sterilized soil. Ten seed of variety L-550 were sown in each pot. Observations on disease incidence were also recorded.

3.2.8.3 Drenching of mycelium suspension

Seven day old seedlings were soil drenched with 100 ml mycelial suspension per pot. Potato dextrose broth was prepared and 100 ml of broth was placed in 250 ml flasks. Thereafter they were autoclaved at 121°C temperature and 15 lbs pressure for 20 minutes. The broth was inoculated with fungus and incubated for 5 days at 25°C. Thereafter the mycelial mats were removed from flasks at the end of the incubation period. Two mycelia mats were macerated in a warring blender with 100 ml sterilized distilled water and incubated in green house for 7 days at 25±2°C.

3.2.8.4 Root dip technique and Fine root cut mycelial suspension dip technique

Seeds of chickpea (variety L 550) surface sterilized with 2.5 per cent sodium hypo-chlorite were sown in autoclaved riverbed sand in pots. The mycelial suspension was prepared as explained earlier. Thereafter the mycelial suspension was transferred into bicker from flasks at the end of the incubation period. The five days old seedling were uprooted and the root system was washed in running water followed by rinsing in sterilized distilled water, whereas some seedling root were cut and then dipped in inoculum solution for 30 seconds. The inoculated seedlings were placed in pots. The pots were moistened with sterile distilled water every alternate day. After 2 days the seedlings were examined for the extent of root damage and for the mortality. Untreated seedlings were served as check (Plate-1).

3.3 Statistical analysis

Statistical analysis of the data was done. Randomized Block Design (RBD) was adopted for field experiments and completely Randomized Design (CRD) was adopted for *in vitro* evaluation and critical difference was worked out at 5% probability level. The values with angular transformation were used wherever necessary.

CHAPTER IV

EXPERIMENTAL FINDINGS

The findings on dry root rot of chickpea with respect to screening of chickpea genotypes, the influence of soil moisture and organic matter on development of dry root rot and validation of inoculation techniques for disease development have been presented here under.

4.1 Collection of diseased samples

Dry root rot infected plants of chickpea were collected from farm field sowing typical diseased symptom located at College of Agriculture, Indore (M.P.), during rabi 2017.

4.2 Symptomatology

Symptoms of dry root rot disease were observed in chickpea plants under natural field condition (Plate-2). Dying plants were observed without showing chlorosis of the whole plants scattered all over the field. The infected plants showed drooping of petiole and leaflet and ultimately drying of whole plant. In some if the infected plant showed whole plant drying along with the topmost leaves remain chlorotic. The leaves of affected plants were straw coloured, but in some cases the lower leaves and stem were also observed brown in colour. The prominent symptoms were found at the flowering and pre-podding stages. Dry plants were seen more in flowering and pre-podding stages, in comparison to seedling stages. After uprooting blackening of the tap root with signs of extensive rotting and loss of lateral and finer roots were observed as characteristic symptoms. The dead roots were quite brittle and accompanied with shredding of bark and lateral roots. In addition of that infected plant uprooting could be performed very easily from the soil.

4.3 Isolation and Identification of associated fungi

The incidence of dry root rot was mostly observed during flowering to podding stage of the crop. The first symptom of the disease is yellowing of the leaves within a day or two such leaves droop and while plant die off in the next two or three days. The leaves and stem of the affected plants were straw coloured. The tap root was dark and quite brittle in dry soil. The dark black sclerotia mycelial aggregation observed on and within host tissue of stem. Dry root rot affected plants were pulled



Plate-2a: View of chickpea plants affected by dry root rot caused by *Rhizoctonia bataticola*



Plate-2b: Blackening showed by dry root rot affected chickpea plant root



Plate-3a: Radial growth of *Rhizoctonia bataticola* on PDA

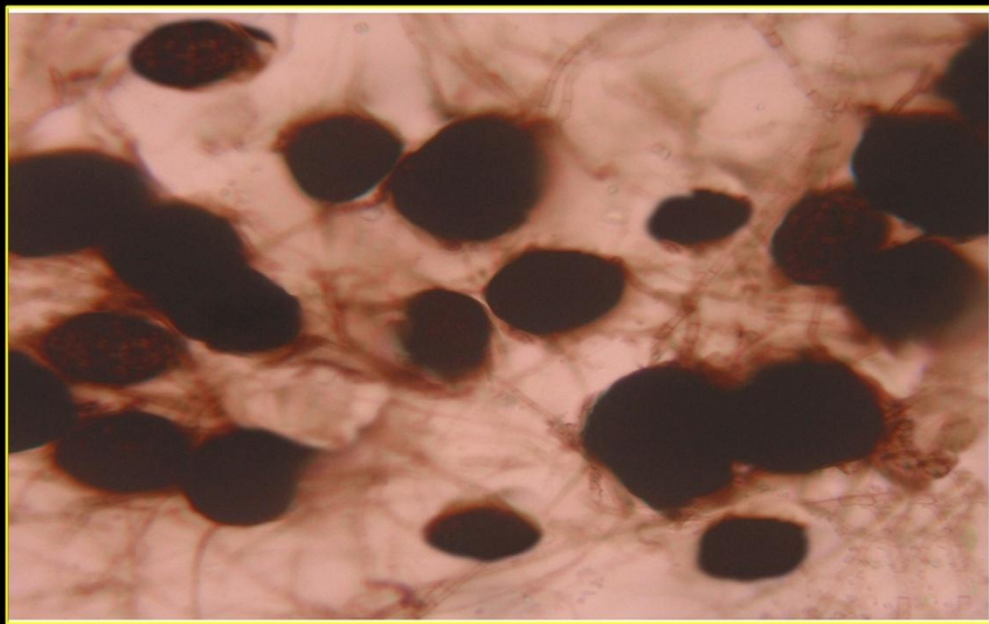


Plate-3b: Sclerotia of *Rhizoctonia bataticola*



Plate-4: Pathogenicity of dry rot by pot inoculation method

bulked easily and showed blackening of roots (Plate-2). The tissues were weakened and break off easily. In advanced cases sclerotial bodies were seen scattered on the affected tissue of root. Isolation made from roots of infected chickpea samples yielded growth of *Rhizoctonia bataticola*. The *Rhizoctonia* culture were purified by hyphal tip method, sub cultured and maintained on PDA slants for further studies as per the technical programme of the thesis. The isolate from roots of infected chickpea plants were identified as *R. bataticola* on the basis of morphological characters, viz., radial hyaline colonies which later became carbonaceous brown to black, mycelium septate and dark brown in colour, typical right angled branching of mycelium and production of sclerotia (Plate-3).

4.4 Pathogenicity test

Pathogenic behavior of *R. bataticola* was tested by sick pot inoculation method. In this method symptoms was initially expressed as drying without chlorosis of the whole plant. Drooping of petioles and leaflet was confined to those at very top of the plant was found. Sometimes when the rest of the plant was dry, the topmost leaves are chlorotic. The leaves of affected plants were straw colored, but in some cases the lower leaves and stem were also observed brown in colour (Plate-4). **On re-isolation, the culture of *Rhizoctonia bataticola* was obtained conforming the Koch's postulates.**

4.5 Screening of chickpea genotypes against dry root rot

Screening of advanced chickpea lines along with popular varieties for resistance to dry root rot in the sick pots maintained at College of Agriculture, Indore (M.P.) during *Rabi*, 2017-18 as described in material and methods. Twenty available promising lines and genotypes of chickpea were screened in sick soil by following standard described techniques with three replication for resistance to dry root rot.

4.5.1 Germination per cent

The data on germination percent of chickpea in sick pots against dry root rot are presented in table-4.1. Significant differences occurred with respect to germination per cent in all varieties/lines.

The germination ranged between 66.67 to 96.67%. Significantly highest germination per cent per was recorded in Vishal (96.67%) which was at par with JG-130 (93.33%), JG-315 (91.67%), JG-62 (90.00%) and RVSSG-27 (88.67%) and followed by GPVT-IG-226, GPVT-JG-218, GPVT-RVJ-201, GPVT-IG-2015-1 and GPVT-J-1140 (80.00, 77.00, 76.67, 75.00 and 70.00%, respectively) as compared to other varieties. Lowest germination was observed in L-550 (66.67%) (Fig.-1).

Table-4.1: Germination per cent of different chickpea variety/lines

Variety/Lines	Germination per cent*
Vishal	96.67 (83.86)
JG-315	91.67 (73.40)
JG-62	90.00 (75.00)
JG-16	67.33 (55.41)
JG-130	93.33 (77.71)
GPVT-JG-218	76.67 (61.71)
GPVT-IG-2015-1	75.00 (60.20)
GPVT-IG-226	80.00 (63.93)
GPVT-J-1140	70.00 (57.00)
RVSSG-27	88.67 (70.94)
GPVT-RVJ-201	77.00 (61.51)
L-550	66.67 (55.33)
SE(m)±	5.30
C.D. at 5%	15.54

*Mean of three replications

**Data in parentheses are $\sqrt{\text{per cent arcsin transformed values}}$

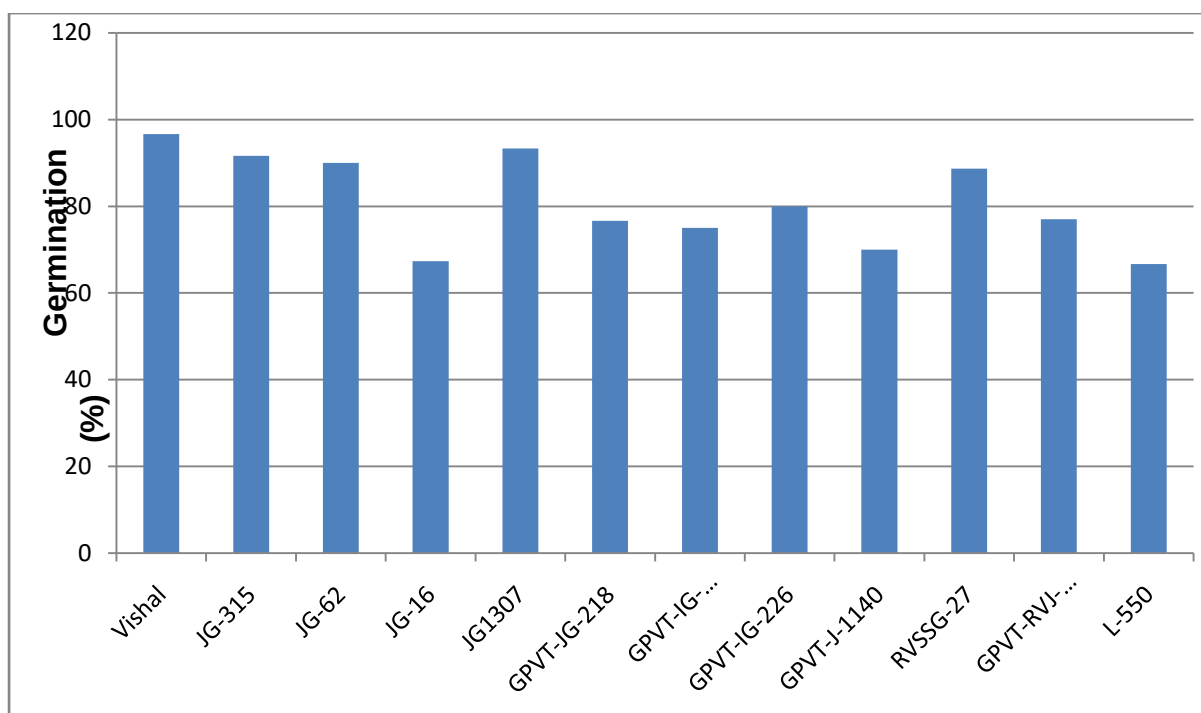


Fig.-1: Germination per cent of different chickpea varieties

Table-4.2: Screening of different genotypes of chickpea for dry root rot in the field (2017-18 Rabi)

Variety	Disease incidence (%) at time*	
	First	Second
Vishal	8.91 (17.16)	10.56 (18.74)
JG-315	1.57 (7.19)	3.50 (10.70)
JG-62	9.12 (17.46)	11.30 (19.52)
JG-16	2.26 (8.51)	4.20 (11.65)
JG-130	6.31 (14.12)	7.40 (15.32)
GPVT-JG-218	3.89 (11.02)	5.56 (13.51)
GPVT-IG-2015-1	1.80 (7.67)	2.12 (8.11)

GPVT-IG-226	1.72 (7.47)	2.78 (9.51)
GPVT-J-1140	3.17 (10.16)	6.35 (14.55)
RVSSG-27	5.40 (13.33)	8.75 (16.99)
GPVT-RVJ-201	8.55 (16.88)	8.19 (15.70)
L-550	10.28 (18.70)	12.00 (20.17)
SE(m)±	1.32	1.71
C.D. at 5%	3.87	5.01

*Mean of three replications

**Data in parentheses are $\sqrt{\text{per cent arcsin transformed values}}$

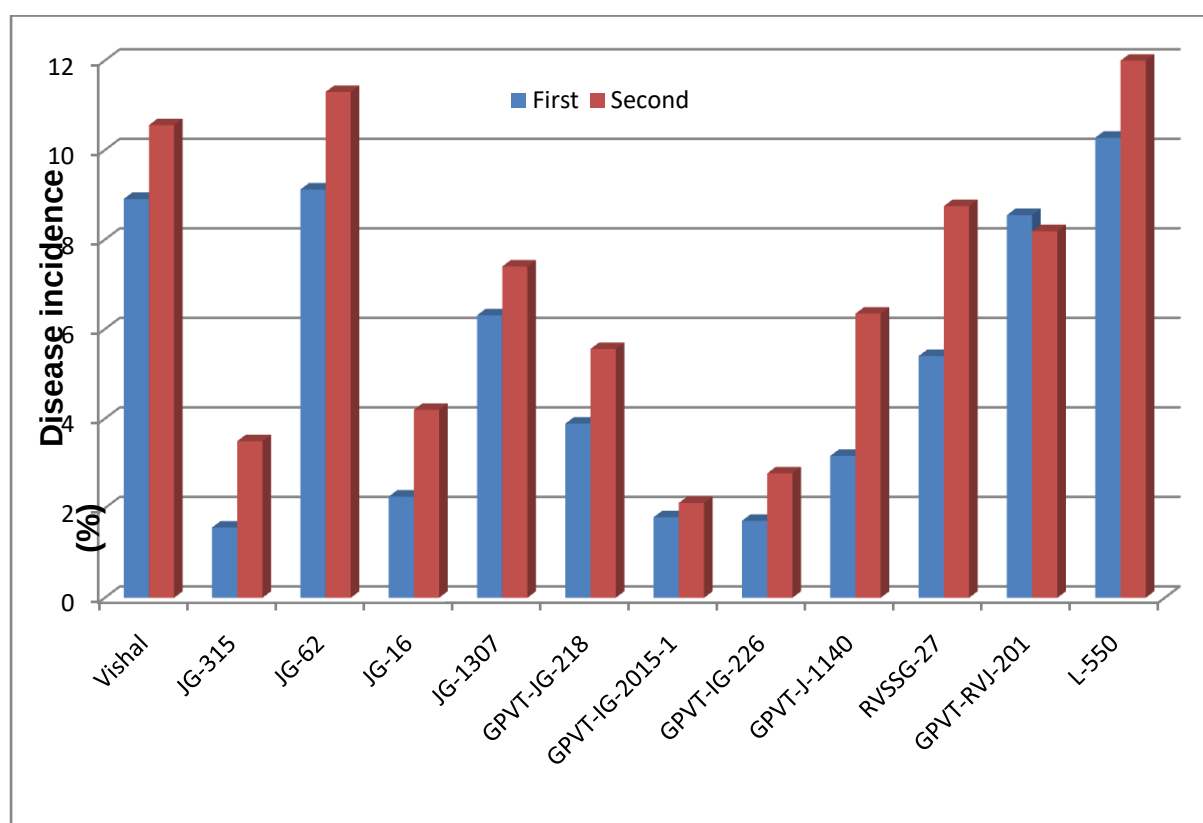


Fig.-2: Screening of different varieties of chickpea for dry root rot in the field (2017-18 Rabi)

4.5.2 Disease incidence

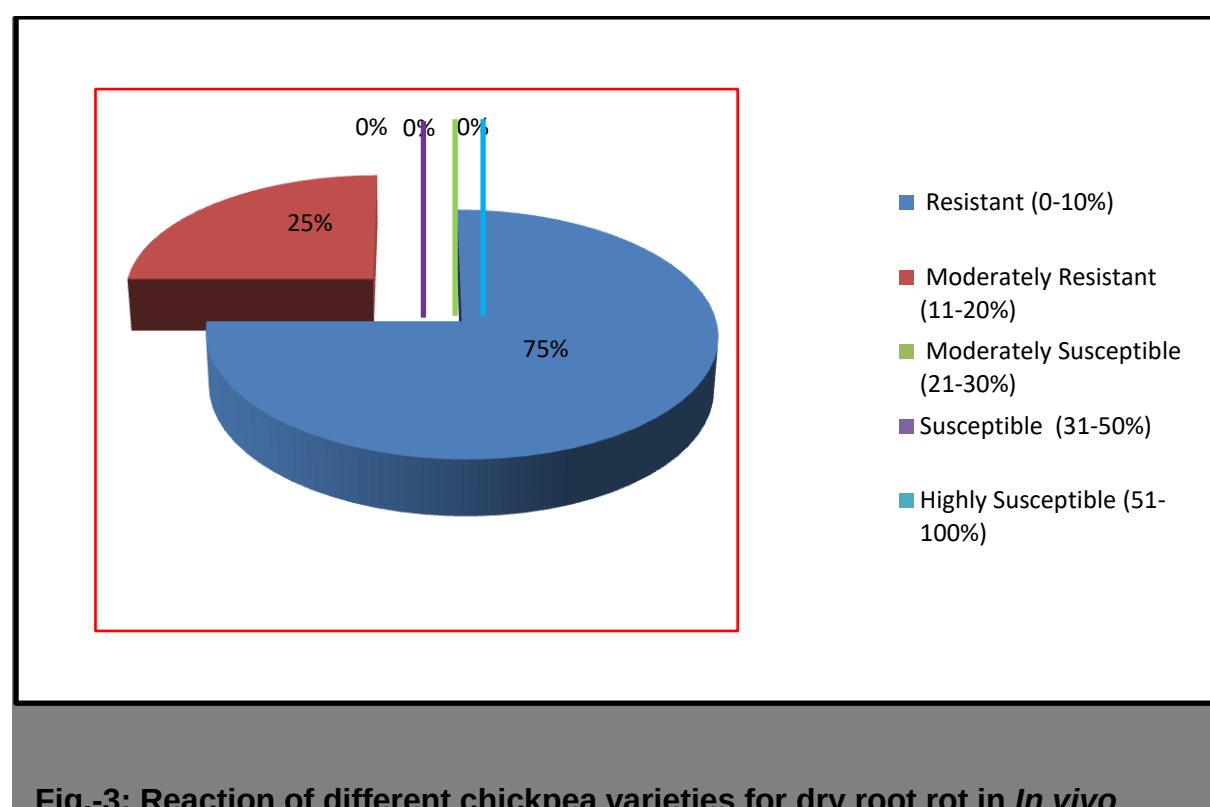
In first observation, significant variation in the dry root rot incidence was observed among the twelve cultivars/lines of chickpea (Table-4.2 and Fig.-2). Incidence of dry root rot caused by *R. bataticola* was significantly less with 1.57% incidence in JG-315 which was at par with GPVT-IG-2015-1 (1.72%), GPVT-IG-226 (1.80%), JG-16 (2.26%), GPVT-J-1140 (3.17%), GPVT-JG-218 (3.89%) and RVSSG-27 (5.40%) as compared to other varieties (Table-4.2). Maximum disease incidence was recorded in genotype L-550 (10.28%) which was at par with JG-62 (9.12%), Vishal (8.91%) and GPVT-RVJ-201 (8.55%). Observations of low disease incidence may be due to less inoculum load persist in the pots and for survival of plants maintenance of higher soil moisture in pots. The outbreak of dry root rot more or less depends on the availability of least soil moisture under natural field condition.

Table-4.2a: Reaction of different chickpea genotypes for dry root rot under pots conditions

S. No.	Reaction	Variety	Number of varieties
1	Resistant (0-10%)	JG-315, JG-16, JG-130, GPVT-RVJ-201, RVSSG-27, GPVT-JG-218, GPVT-IG-2015-1, GPVT-IG-226 and GPVT-J-1140	9
2	Moderately Resistant (11-20%)	Vishal, JG-62 and L-550	3
3	Moderately Susceptible (21-30%)	-	0
4	Susceptible (31-50%)	-	0
5	Highly Susceptible (51-100%)	-	0

In second time observation, GPVT-IG-2015-1 was the best variety, which showed less incidence (2.12%) against the dry root rot of chickpea which was at par with GPVT-IG-226 (2.78%), JG-315 (3.50%), JG-16 (4.20%), GPVT-JG-218 (5.56%) and GPVT-J-1140 (6.35%) as compared to other cultivars (Table-4.2). Maximum disease incidence was recorded in variety L-550 (12.00%) (Fig.-2).

The results obtained are presented in table-4.2a and fig.-3. Among twelve varieties/lines screened against dry rootrot, nine of the varieties showed resistant reaction viz., JG-315, JG-16, JG-130, GPVT-RVJ-201, RVSSG-27, GPVT-JG-218, GPVT-IG-2015-1, GPVT-IG-226 and GPVT-J-1140, whereas three varieties viz., Vishal, JG-62 and L-550 showed moderately resistant reactions with 11-20 per cent disease incidence, none of the variety showed moderately susceptible reaction, susceptible reaction and highly susceptible reaction.



4.6 Effect of soil moisture and organic matter content

4.6.1 On germination per cent with variety JAKI-9218

The data presented in table-4.3 indicate that significant differences occurred with respect to germination in soil moisture and organic matter content treatments with variety JAKI-9218.

In *R. bataticola* infested soil, germination ranged between 35.00 to 83.33 percent. Significantly highest germination per cent was observed in high organic matter with 200 ml water/pot and medium organic matter with 150 ml water/pot and low organic matter with 100 ml water/pot were showed lowest germination with equal per cent (35.00%) (Table-4.3 and Fig.-4).

The germination ranged between 83.33 to 91.67% in *R. bataticola* uninfested soil. Significantly highest germination per cent was recorded in high organic matter with 200 ml water/pot (91.67%) which was at par with medium organic matter with 150 ml water/pot and low organic matter with 100 ml water/pot (83.33%).

Table-4.3: Germination percentage of chickpea in *R. bataticola* infested and uninfested soil in different soil moisture and organic matter contents with JAKI-9218 varieties

Treatment	Germination (%)*		
	JAKI-9218		
	Infested soil	Uninfested soil	Mean
100 ml water/ potwith 1.74% OM	35.00 (36.19)**	83.33 (69.14)**	59.17 (52.66)**
150 ml water/ pot with 1.88% OM	83.33 (69.14)	83.34 (72.37)	83.33 (70.75)
200 ml water/ pot with 1.95% OM	83.33 (72.73)	91.67 (77.95)	87.50 (75.34)
MEAN	67.22 (59.35)	86.11 (73.15)	
	Treatment (T)	Soil type (S)	Inter (T×S)
SE(m)±	4.63	5.67	8.02
C.D. at 5 %	13.51	16.55	23.40

*Mean of three replications

**Data in parentheses are arcsin transformed values

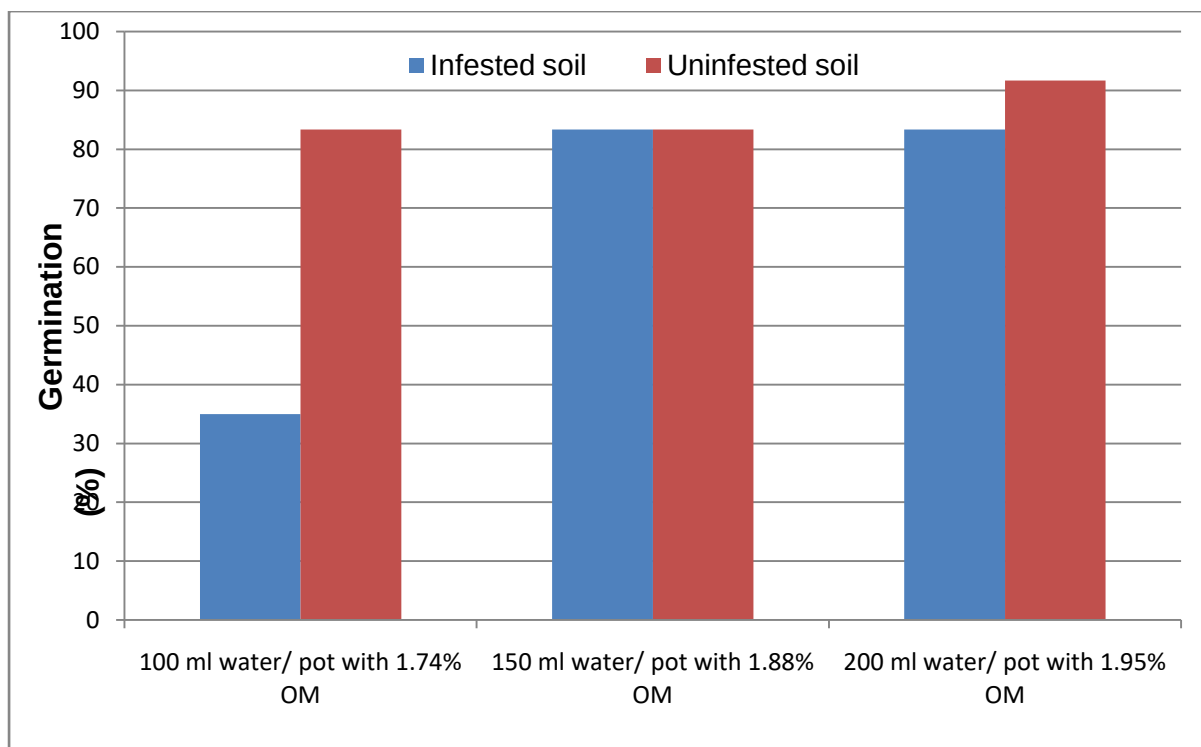


Fig.-4: Germination percentage of chickpea in *R. bataticola* infested and uninfested soil in different soil moisture and organic matter contents with JAKI-9218 variety

4.6.2 On disease incidence per cent with variety JAKI-9218

Data in the table-4.5 and plate-5 indicate that the first observation at 10 days after sowing (DAS) of incidence of dry root rot caused by *R. bataticola* was significantly less in low organic matter with 100 ml water/pot which was at par with medium organic matter with 150 ml water/pot with variety JAKI-9218.

At 20 DAS, 100 ml water/pot with 1.74% OM showed minimum incidence per cent and maximum incidence per cent (57.08%) was recorded in 200 ml water/pot with 1.95% OM which was at par with 150 ml water/pot with 1.88% OM (55.83%).

Table-4.4: Dry root rot incidence of chickpea in *R.bataticola* infested soil in different soil moisture and organic matter contents with JAKI-9218 variety

Treatment	Observations of disease incidence (%) at*			
	10 DAS	20 DAS	30 DAS	Mean
100 ml water/ pot with 1.74% OM	0.00 (0.00)**	20.83 (15.00)**	91.67 (90.00)**	37.50 (35.00)**
150 ml water/ pot with 1.88% OM	20.84 (8.03)	55.83 (39.83)	85.00 (73.08)	53.89 (40.31)
200 ml water/ pot with 1.95% OM	36.25 (40.00)	57.08 (61.75)	66.25 (65.00)	53.19 (55.58)
MEAN	19.03 (16.01)	44.58 (38.86)	80.97 (76.03)	
		Treatment (T)	Observation (O)	Inter (T×O)
SE(m)±		9.46	9.46	16.38
C.D. at 5 %		27.60	27.60	47.81

*Mean of three replications

**Data in parentheses are arcsin transformed values

200 ml water/pot with 1.95% OM was showed lowest dry root rot incidence at 30 DAS, where 100 ml water/pot with 1.74% OM was expressed the incidence of root rot was 91.67 percent.



Plate-5: Dry root rot incidence percentage of chickpea in *Rhizoctonia bataticola* infested soil in different soil moisture and organic matter contents with JAKI-9218 variety

4.7 Inoculation techniques to induce dry root rot disease

Different artificial techniques viz., Seed inoculation method, Pot inoculation method or sick soil method. Drenching of mycelium suspension and mycelia disc at the base of plant, Root dip technique, Fine root cut mycelia suspension dip technique were evaluated to inoculation of *Rhizoctonia bataticola*. The data presented in table-4.5 and plate-6 indicates the effectiveness of inoculation techniques with disease incidence per cent.

Table-4.5: Incidence of dry root rot recorded by different artificial inoculation techniques

Artificial inoculation techniques	Dry root rot incidence (%)*
Seed inoculation method	76.38 (60.96)**
Pot inoculation method	75.43 (60.31)
Drenching of mycelium suspension	80.96 (64.18)
Root dip technique	83.78 (66.52)
Fine root cut mycelia suspension dip technique	84.87 (67.36)
SE(m)±	1.63
C.D. at 5 %	4.62

*Mean of three replications

**Data in parentheses are $\sqrt{\text{per cent arcsin transformed values}}$

Fine root cut mycelia suspension dip technique was the best artificial technique with the highest incidence of dry root rot(84.87%)which was at par with root dip technique and drenching of mycelium suspension (83.78 and 80.96%, respectively). The lowest incidence was recorded in pot inoculation technique (75.43%). Initial mechanical injuries in fine root cut mycelia suspension technique may provide primary passage of tissue injuries in fine roots, which leads to comparatively quick dry root rot infection.



**Seed inoculation
method**



**Pot inoculation
method**



**Drenching of mycelium
suspension**



Root dip technique



**Fine root cut mycelia
suspension dip technique**

Plate-6: Incidence of dry root rot in different artificial inoculation techniques

CHAPTER V

DISCUSSION

Results obtained from the investigation on “studies on dry root rot [*Rhizoctonia bataticola*(Taub.) Butler] of chickpea.” stated in chapter IV have been examined critically and discussed here with appropriate interpretations.

Dry root rot infected plants of chickpea were collected from diseased field located at College of Agriculture, Indore (M.P.), during 2017.

Symptoms of dry root rot disease were observed in chickpea under natural field condition. Wilting without chlorosis of the whole plant scattered all over the field was observed. Drooping of petioles and leaflet was confined to those at very top of the plant was found. Sometimes when the rest of the plant was dry, the topmost leaves are chlorotic. The leaves of affected plants were straw colored, but in some cases the lower leaves and stem were brown. The prominent symptoms were found at the flowering and pre-podding stage. Wilted plants were seen more in flowering and pre-podding stages, in comparison to seedling stages. After uprooting blackening of the tap root with signs of extensive rotting and loss of lateral and finer roots were observed as characteristic symptoms. The dead roots were quite brittle and accompanied with shredding of bark and lateral roots. Similar symptoms were observed by Sharma *et al.*, (2015).

The pathogen isolated from the root region was purified by single hyphal tip method and maintained on PDA for further studies. **On the basis of morphological characters and pathogenicity test the fungus responsible for dry rot of chickpea was identified as *Rhizoctonia bataticola* (*Macrophomina phaseolina*).**

On PDA at 25°C the culture of *R. bataticola* was brown to black colour with fluffy growth. The mycelium was pale white in colour in the initial stages of the growth but later turned to dark brown to black as and when sclerotial formation started. Mycelia showed right angled branching. The sclerotia varied in shape and size with growth pattern of scattered to clustered. These observations are in line with the description provided by Butler for Pycnidial stage *M. phaseolina* and other characters described by Ashby (1927) and Manjunatha (2011).

Based on morphological characters the pathogen was identified as *R. bataticola* Butler. Similar reports were observed by Barnett and Hunter (1972) and Sajana *et al.* (2004).

Pathogenic behavior of *R. bataticola* was established by pot inoculation method. In this method symptoms were initially expressed as wilting without chlorosis of the whole plant scattered all over the field was observed. Drooping of petioles and leaflet was confined to those at very top of the plant was found. Sometimes when the rest of the plant was dry, the topmost leaves are chlorotic. The leaves of affected plants were straw colored, but in some cases the lower leaves and stem were also observed brown colour. **On re-isolation, the culture of *Rhizoctonia bataticola* was obtained conforming the Koch's postulates. Similar symptoms were observed by** Amrutha *et al.* (2014).

Twelve available promising cultivars of chickpea were screened in sick soil by standard techniques for resistance to dry root rot at College of Agriculture, Indore (M. P.) during Rabi, 2017-18.

Significant differences occurred with respect to germination per cent in all genotypes. The germination ranged between 66.67 to 96.67%. Significantly highest germination per cent was recorded in Vishal which was at par with JG-1307, JG-315, JG-62 and RVSSG-27 and followed by GPVT-IG-226, GPVT-JG-218, GPVT-RVJ-201, GPVT-IG-2015-1 and GPVT-J-1140 as compared to other varieties. Lowest germination was observed in L-550.

In first time observation, incidence of dry root rot caused by *R. bataticola* was significantly less incidence in JG-315 which was at par with GPVT-IG-2015-1, GPVT-IG-226, JG-16, GPVT-J-1140, GPVT-JG-218 and RVSSG-27 as compared to other varieties. Maximum disease incidence was recorded in genotype L-550 which was at par with JG-62, Vishal and GPVT-RVJ-201.

In second time observation, GPVT-IG-2015-1 was the best variety, which showed less incidence in controlling dry root rot of chickpea which was at par with GPVT-IG-226, JG-315, JG-16, GPVT-JG-218 and GPVT-J-1140 as compared to other cultivars. Maximum disease incidence was recorded in variety L-550. The incidence of dry root rot incidence was quite low in screening experiment, that is the reason none of the lines were showed high level of disease incidence.

Among twelve varieties screened for dry rootrot, nine of the varieties showed resistant reaction viz., JG-315, JG-16, JG-1307, GPVT-RVJ-201, RVSSG-27, GPVT-JG-218, GPVT-IG-2015-1, GPVT-IG-226 and GPVT-J-1140, whereas three varieties viz., Vishal, JG-62 and L-550 showed moderately resistant reactions with 11-20 per cent disease incidence. Pathogenic variation in *R. bataticola* have also been reported from different parts of country (Lekhraj *et al.*, 2012, Sharma *et al.*, 2012), which supports the present findings from Madhya Pradesh.

Significant differences occurred with respect to germination per cent in soil moisture and organic matter content treatments with variety JAKI-9218. In *R. bataticola* infested soil, germination ranged between 35.00 to 83.33 percent. Significantly highest germination per cent was observed in 200 ml water/pot with 1.95% OM and 150 ml water/pot with 1.88% OM and 100 ml water/pot with 1.74% OM were showed lowest germination.

The germination ranged between 83.33 to 91.67% in *R. bataticola* uninfested soil. Significantly highest germination per cent was recorded in 200 ml water/pot with 1.95% OM which was at par with 150 ml water/pot with 1.88% OM and 100 ml water/pot with 1.74% OM.

The first observation of incidence of dry root rot caused by *R. bataticola* was significantly less in 100 ml water/pot with 1.74% OM which was at par with 150 ml water/pot with 1.88% OM with variety JAKI-9218.

In second observation, low organic matter with 100 ml water/pot was showed minimum incidence per cent and maximum incidence per cent was recorded in 200 ml water/pot with 1.95% OM which was at par with 150 ml water/pot with 1.88% OM. 200 ml water/pot with 1.95% OM was showed lowest dry root rot incidence in third observation, where 100 ml water/pot with 1.74% OM. Role of soil moisture on disease severity of dry root rot in chickpea was also studied by Sharma and Pande (2013) under controlled environment.

Different artificial techniques viz., Seed inoculation method, Pot inoculation method or sick soil method. Drenching of mycelium suspension and mycelia disc at the base of plant, Root dip technique, Fine root cut mycelia suspension dip technique were evaluated to inoculation of *R. bataticola*. Fine root cut mycelia suspension dip technique was the best artificial technique with the highest incidence

of dry root rot which was at par with root dip technique and drenching of mycelium suspension. The lowest incidence was recorded in pot inoculation technique. Pande *et al.* (2012) also evaluated the different techniques for artificial inoculation of dry root rot of chickpea.

Mehta, (2004) tested different inoculation techniques were for creating maximum disease leaf blight of mungbean caused by *Macrophomina phaseolina* (Tassi) Goid.

Similarly Sharma *et al.* (2013) reported of unraveling effects of temperatures and soil moisture stress responses on development of dry root rot (*R. bataticola*) in chickpea. Effect of soil moisture regimes, *i.e.* 40%, 60%, 80% and 100% was studied on development of dry root rot under controlled environment.

CHAPTER-VI

SUMMARY, CONCLUSION AND SUGGESTION FOR FURTHER WORK

❖ Summary

The present investigations on dry root rot [*Rhizoctonia bataticola* (Taub.) Butler] of chickpea was carried out during *rabi* 2017-18, at Department of Plant Pathology and field of College of Agriculture, Indore (M.P.), R.V.S.K.V.V. Gwalior (M.P.)

- ✓ The dry root rot symptoms observed were mainly brittleness of tap root and presence of minute sclerotia. These symptoms separated this disease from the wilt complex as termed by various workers in India.
- ✓ The pathogen isolated from the root region was purified by single hyphal tip method and maintained on PDA for further studies. Based on morphological characters the pathogen was identified as *R. bataticola* Butler.
- ✓ Pathogenicity of the fungus was established by inoculation and re-isolation from artificially infected plants. In the present study, soil inoculation method was followed to establish the disease. Soil inoculation method as the most suitable in establishing the disease caused by *R. bataticola*.
- ✓ Twelve available promising cultivars of chickpea were screened in sick soil by standard techniques for resistance to dry root rot. Among twelve varieties screened for dry root rot, nine of the varieties showed resistant reaction viz., JG-315, JG-16, JG-130, GPVT-RVJ-201, RVSSG-27, GPVT-JG-218, GPVT-IG-2015-1, GPVT-IG-226 and GPVT-J-1140, whereas three varieties viz., Vishal, JG-62 and L-550 showed moderately resistant reactions with 11-20 per cent disease incidence.
- ✓ In *R. bataticola* infested soil, significantly highest germination per cent was observed in 200 ml water/pot with 1.95% OM and 150 ml water/pot with 1.88% OM and 100 ml water/pot with 1.74% OM were showed lowest germination in JAKI-9218.
- ✓ In *R. bataticola* uninfested soil, significantly highest germination per cent was recorded in 200 ml water/pot with 1.95% OM which was at par with 150 ml water/pot with 1.88% OM and 100 ml water/pot with 1.74% OM in JAKI-9218.

- ✓ **The first observation of** incidence of dry root rot caused by *R. bataticola* was significantly less in 100 ml water/pot with 1.74% OM which was at par with 150 ml water/pot with 1.88% OM with variety JAKI-9218. In second observation, 100 ml water/pot with 1.74% OM was showed minimum incidence per cent and maximum incidence per cent was recorded in 200 ml water/pot with 1.95% OM which was at par with 150 ml water/pot with 1.88% OM. 200 ml water/pot with 1.95% OM was showed lowest dry root rot incidence in third observation, where 100 ml water/pot with 1.74% OM.
- ✓ Fine root cut mycelia suspension dip technique was the best artificial technique with the highest incidence of dry root rot which was at par with root dip technique and drenching of mycelium suspension. The lowest incidence was recorded in pot inoculation technique.

❖ **Conclusion**

- ✓ Among twelve varieties screened for dry root rot, nine of the varieties showed resistant reaction viz., JG-315, JG-16, JG-1307, GPVT-RVJ-201, RVSSG-27, GPVT-JG-218, GPVT-IG-2015-1, GPVT-IG-226 and GPVT-J-1140.
- ✓ In *R. bataticola* infested soil and uninfested soil, significantly highest germination per cent was observed in 200 ml water/pot with 1.74% OM in variety JAKI-9218.
- ✓ **The observation on disease** incidence of dry root rot caused by *R. bataticola* on different organic matter and moisture regime showed that low soil moisture content followed by low level of organic matter (1.74%) few our the dry root rot incidence.

❖ **Suggestion for further work**

- Confirmation of genetic potential of resistant lines of chickpea should be repeated in upcoming season by insuring optimum inoculum road.
- Standardization the technique for maintenance of accurate soil moisture for whole life cycle of plants during experimentation.
- Exploit the probability of sophisticate instruments for exact the quantification of soil moisture during the life cycle of chickpea plants.
- Further authentication of inoculation technique for *in vitro* evaluation of different inoculation techniques.

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APPENDICES

Analysis of Variance Table – 4.1

	DF	SS	MSS	Fcal	Ftab
Rep	2	14.46	7.23	0.09	3.44
Treat	11	2995.08	272.28	3.23	2.26
Error	22	1853.93	84.27		
Total	35	4863.46			

Analysis of Variance Table – 4.2 (First)

	DF	SS	MSS	Fcal	Ftab
Rep	2	32.53	16.26	3.11	3.44
Treat	11	622.82	56.62	10.83	2.26
Error	22	115.00	5.23		
Total	35	770.35			

Analysis of Variance Table – 4.2 (Second)

	DF	SS	MSS	Fcal	Ftab
Rep	2	31.51	15.76	1.35	3.44
Treat	11	518.54	47.14	4.03	2.26
Error	22	257.12	11.69		
Total	35	807.17			

Analysis of Variance Table – 4.3 (JAKI-9218)

SV	DF	SS	MSS	FCAL	FTAB
Treatment (T)	1	1142.68	1142.68	4.44	4.41
Soil type (S)	2	2299.45	1149.72	4.47	3.55
Inter(T×S)	2	106446.46	53223.23	206.94	3.55
Error	18	4629.48	257.19		
TOTAL	23	114518.07	55772.83		

Analysis of Variance Table – 4.4 (JAKI-9218)

SV	DF	SS	MSS	FCAL	FTAB
Treatment (T)	2	22024.59	11012.30	10.26	3.35
Observation (O)	2	11179.72	5589.86	5.21	3.35
Inter(T×O)	4	22944.21	5736.05	5.35	2.73
Error	27	28972.26	1073.05		
TOTAL	35	85120.78	23411.25		

Analysis of Variance Table – 4.5

SV	DF	SS	MSS	FCAL	FTAB
Treatment	4	161.72	40.43	3.79	3.06
Error	15	160.05	10.67		
TOTAL	19	321.78	51.10		

VITA

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